

Technical artist and graphics programmer who works across code, shaders, VFX, UI, and art integration. I directly build C++/OpenGL engine features and Unity visual systems, diagnose rendering and gameplay failures, and translate visual goals into testable technical requirements without obscuring authorship boundaries.

SELECTED EXPERIENCE

TEACHING ASSISTANT - GAME DEVELOPMENT PROJECT I

DIGIPEN KOREA | SPRING 2025

- Supported students in DigiPen's Korea program with C++ implementation, debugging, and technical problem-solving during game project development.
- Reviewed student projects and provided clear, actionable technical feedback to help teams identify issues and improve their implementations.

MANZO

C++ / OpenGL / GLSL | 2024-2025

- Implemented BPM timing windows, beat/bar counting, and audio-synchronized player movement and boss patterns in a custom C++ engine.
- Built layer-sorted draw queues, framebuffer post-processing for bloom, underwater distortion, god rays, ripples, and transitions, plus particles with linear, curved, radial, spray, random, and player-targeted motion.
- Moved scenario/dialogue ownership into engine-level systems to eliminate dangling-pointer failures; traced severe boss slowdown to redundant collision checks and removed repeated work. Largest repository contributor: 366 commits.

TOO HOT!

Unity / ShaderLab / VFX / UI | 2026

- Created and integrated the game's 2D shadow treatment, pattern-specific VFX, UI, animation, hit feedback, and visual hierarchy; tuned width and length controls for readable shadows across characters and combat spaces.
- Specified GameManager and per-stage ScriptableObject data flow, save-range safeguards, chapter selection, and clean-state debug controls; reviewed and tested teammate-authored gameplay implementations.
- Balanced direct art/technical-art execution with a 130+ item P0-P3 backlog, two-programmer coordination, code review, merges, and final visual integration.

STREET TYPER

Unity URP / UI Technical Art | 10-day build, 2026

- Owned original 2D art, UI composition, particles, outlines, camera shake, hit VFX, and animated feedback for a shipped bilingual typing-combat game.
- Specified, evaluated, debugged, and integrated an AI-assisted reusable UI shader workflow for rounded forms, gradients, drop/inner shadows, blur, presets, and inspector iteration; gameplay code was teammate-authored.
- Published a playable build on itch.io and preparing the game for a Steam release.

ADDITIONAL EVIDENCE

DOUBLE HIT - Implemented C++ texture/sprite management, collision, GameObject/GameComponent architecture, and shared engine services.

BIRD STRIKE - Implemented audio-timeline beat detection, rhythm-synchronized spawning, dynamic attack subdivision, player/crow movement, and atan2 direction logic; also produced original art and audio.

NEW MANZO - Contribute Unity gameplay and technical systems for fish schooling, obstacle avoidance, beat-linked hunting, raycasting, and post-processing; repository lead contributor with 417 commits.

SKILLS

Graphics	OpenGL, GLSL, Unity URP, ShaderLab, framebuffer/post-processing, custom shadows, particles, VFX, UI shaders
Programming	C++, C#, C, Python, JavaScript; gameplay and engine architecture, collision, debugging, memory/lifetime fixes
Workflow	Git branching and merge review, GitHub Projects/Issues, Notion, CMake, Visual Studio, WSL, profiling, technical specification

EDUCATION

B.S. Computer Science in Real-Time Interactive Simulation | 2026-Present | Expected Graduation: May 2028
Keimyung University / DigiPen Institute of Technology | GPA 3.958/4.0